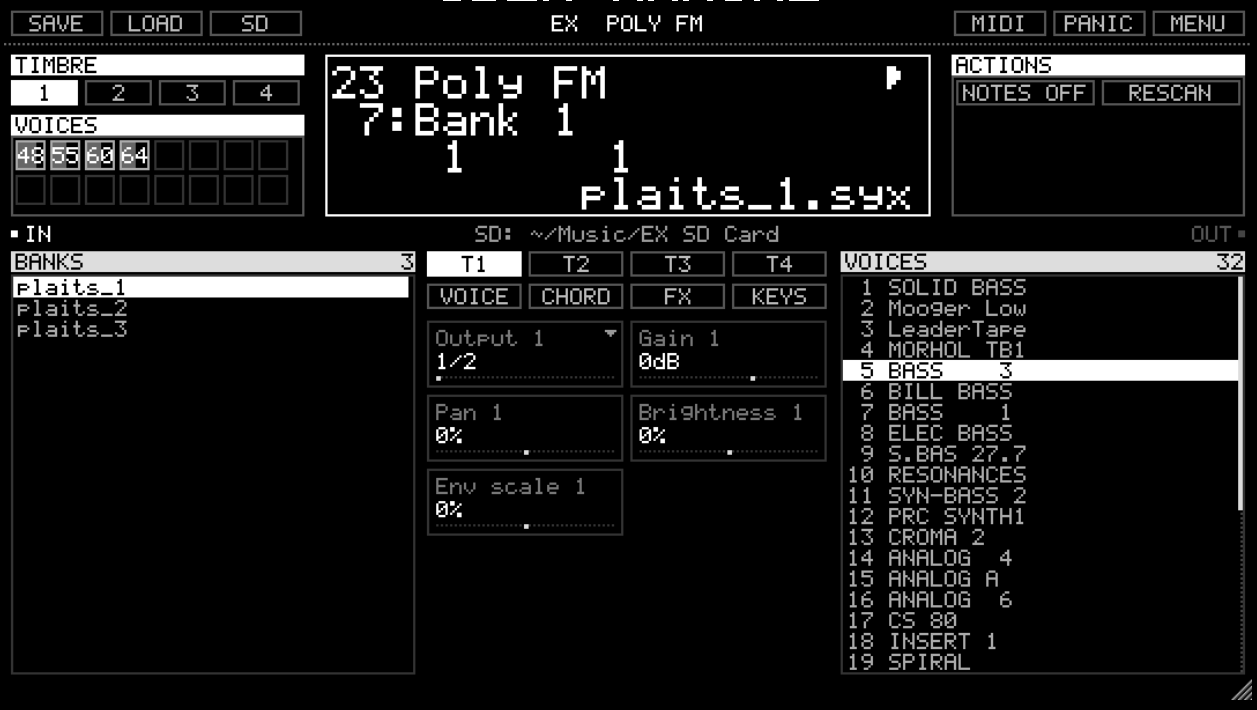
EX-GF

USER MANUAL



Nine disting EX-style instruments and effects: **EX Multisample** · **EX Poly Wavetable** · **EX Poly FM** · **EX Poly Exciter** · **EX Granulator** · **EX Multi-FX** · **EX Resonator** · **EX Spectral Freeze** · **EX Macro Oscillator 2**

VST3 · AU · LV2 · standalone - macOS, Windows and Linux

Version 1.1.0 - github.com/femifleming/ex-gf

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INTRODUCTION

EX-GF is a suite of nine software instruments and effects modelled on algorithms of the Expert Sleepers **disting EX** Eurorack module. They follow the module's manual (firmware 1.26) parameter for parameter, use the same file formats (sample folders, wavetables, exciters, DX7 banks, Scala tunings) and wear the module's look: the whole window is one white-on-black pixel display in the module's own 5×7 font.

Plug-in	Algorithm(s)	In short
EX Multisample	3 - SD Multisample 5 - WAV Recorder	16-voice, 3-timbre sample player with velocity layers, round robins, chords, arpeggiator, EQ and delay; plus a recorder with an **auto-sampler** that samples any MIDI/CV instrument into a ready-to-play folder.
EX Poly Wavetable	10 - Poly Wavetable	16-voice wavetable synth: 2 envelopes, filter, LFO, unison, Juno-style chorus and delay. Imports WaveEdit banks.
EX Poly FM	23 - Poly FM	16-voice, 4-timbre 6-operator FM (the Plaits engine). Loads DX7 .syx banks; DX7 editors can edit it over SysEx.
EX Poly Exciter	13 - Poly Exciter	16-voice exciter/resonator synth: plucked, struck and bowed-like sounds from short exciter samples or live input.
EX Granulator	11 - Granulator	Effect. Granular synthesis on live input or a WAV file, played from MIDI notes or three drones, with delay and reverb.
EX Multi-FX	12 - Multi FX	Effect. 3-band EQ, chorus/flanger/phaser/pitch shift, delay and reverb with free routing and a variable sample rate.
EX Resonator	20 - Resonator	Effect played from MIDI. Mutable Instruments Rings: six resonator models and the string-synth mode.
EX Spectral Freeze	24 - Spectral Freeze	Effect. Captures a moment of sound and sustains it: four phase-vocoder voices with movement and pitch shift.
EX Macro Oscillator 2	21 - Macro Oscillator 2	Four instances of Mutable Instruments Plaits (24 models), as a 4-voice synth or four separate oscillators.

Each instrument is available as a **VST3** (all platforms), **Audio Unit** (macOS), **LV2** (Linux) and a **standalone app**. Settings are saved with your DAW project, and every plug-in also remembers its last state between sessions.

EX-GF is not affiliated with or endorsed by Expert Sleepers; "disting" is their trademark and is used only to describe compatibility. If you own the module, its MicroSD card works directly with these plug-ins.

INSTALLING

All downloads are on the release page: **github.com/femifleming/ex-gf/releases**.

MACOS — HOMEBREW

```
brew tap femifleming/tap
brew trust --tap femifleming/tap
brew install --cask femifleming/tap/ex-gf            # apps + VST3 + AU
brew install --cask femifleming/tap/ex-gf-sd-card    # optional sample library
```

Homebrew 7 asks you to *trust* third-party taps once. To build from source instead (macOS or Linux): `brew install --formula femifleming/tap/ex-gf`, then follow the link instructions it prints.

MACOS — DOWNLOAD

- **EX-GF-1.1.0-macOS.zip**: unzip and double-click **install.command** - apps go to ~/Applications, plug-ins to ~/Library/Audio/Plug-Ins/VST3 and /Components.
- **EX-GF-1.1.0-macOS.pkg**: installs for all users (right-click -> Open the first time).
- The builds are universal (Apple silicon + Intel) but not notarised. If macOS or your DAW refuses to open them, run:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/VST3/EX*.vst3 \
    ~/Library/Audio/Plug-Ins/Components/EX*.component ~/Applications/EX*.app
```

WINDOWS

Unzip **EX-GF-1.1.0-Windows-x64.zip**, right-click **install.ps1** -> *Run with PowerShell* (as administrator). VST3 plug-ins go to C:\Program Files\Common Files\VST3; the apps appear in the Start menu under **EX-GF**.

LINUX

```
tar xzf EX-GF-1.1.0-Linux-x86_64.tar.gz
cd ex-gf-1.1.0-Linux-x86_64 && ./install.sh
```

VST3 -> ~/.vst3, LV2 -> ~/.lv2, apps -> ~/.local/bin with desktop menu entries.

THE SAMPLE LIBRARY

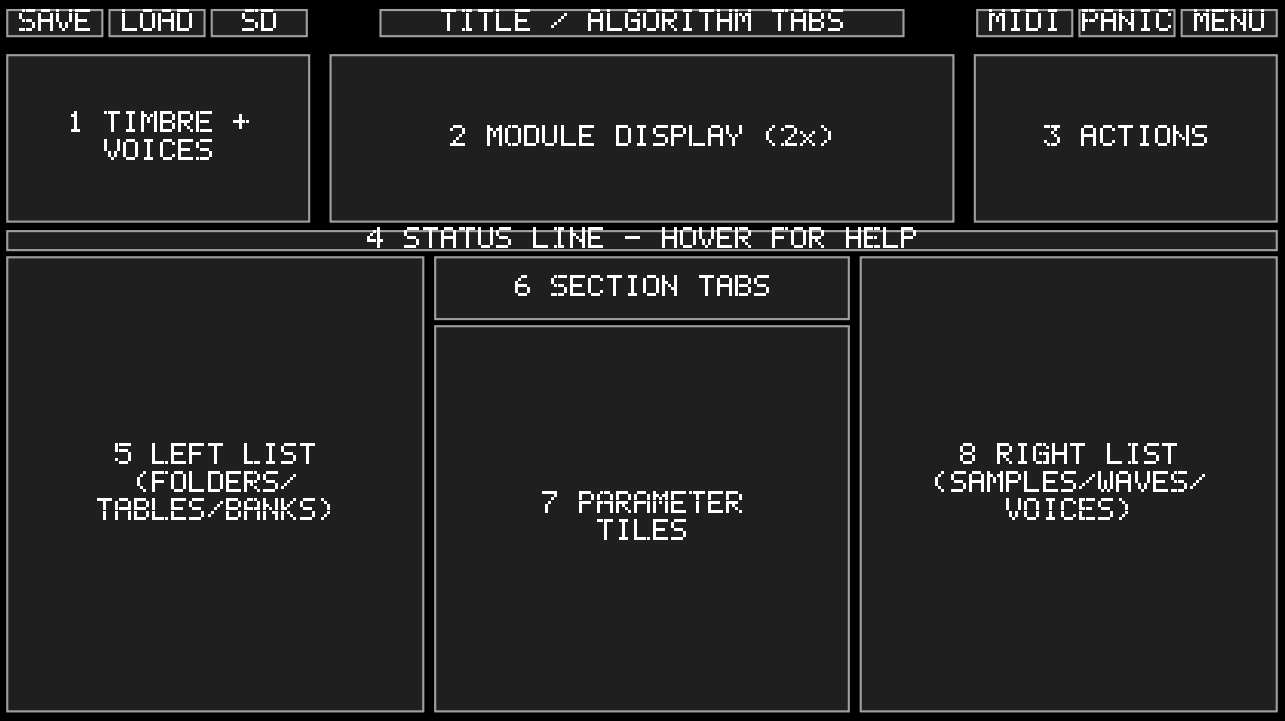
Download **EX-SD-Card.zip** and unzip it into your **Music** folder so you have **Music/EX SD Card** (the default location), or pick any folder with the **SD** button. It holds 18 multisampled instruments, 38 wavetables, 5 exciter tables, DX7 banks and Scala tunings - all free (CC0 / CC-BY / MIT, see CREDITS.txt inside). A real disting EX MicroSD card works too.

BUILDING FROM SOURCE

```
git clone --recursive https://github.com/femifleming/ex-gf
cd ex-gf
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release
```

THE SCREEN

Every plug-in uses the same mirror-symmetrical layout. The window can be resized; it snaps to whole-pixel sizes (1×, 2×, 3× …) so the pixels stay sharp.



	Area	What it does
	Top bar	**SAVE / LOAD** presets (stored on the SD card), **SD** chooses the card folder, **MIDI** channels, **PANIC** (all notes off), **MENU** (every option of the module's menus). In EX Multisample the centre holds the algorithm tabs.
1	Timbre / voices	Timbre, oscillator or target buttons choose what the lists edit. The cells show each voice's note and level (in the effects: input, section or voice levels).
2	Module display	The disting EX's own 128×32 screen: current parameter, live voices, levels, auto-sampler progress. Mouse wheel = parameter, shift+wheel = value; Up/Down and Left/Right act like the P and V encoders, Enter = push P (menu), Esc = push V.
3	Actions	Buttons for the algorithm: RECORD, PLAY, AUTO, TEST, IMPORT, WAVEEDIT, FREEZE, STRUM, BYPASS, NOTES OFF …
4	Status line	Hover anything for a description. MIDI IN / OUT activity lights sit at either end.
5/8	Lists	Click to load: sample folders and their samples, wavetables and their waves, exciter tables and exciters, FM banks and voices, recordings.
6	Section tabs	Pages of related parameters (e.g. ENV, FILTER, CHORD, FX).
7	Parameter tiles	Drag up/down or use the wheel to change (hold shift for fine steps); double-click resets to the default. Tiles with a small (v) open a list of choices when clicked.

PLAYING NOTES

Play from MIDI, or click the window and use the computer keyboard: **A W S E D F T G Y H U J K O L P ;** play a chromatic octave and a half, **Z**/**X** shift the octave.

MIDI

- The receive channel defaults to 1 (Omni is available under MIDI). Multi-timbral plug-ins listen on consecutive channels (timbre 1 on the receive channel, timbre 2 on the next …) unless you assign each timbre its own channel.
- Recognised: note on/off, velocity, pitch bend, sustain (CC64), All Notes Off (CC123); channel and poly aftertouch in Poly Exciter; DX7 SysEx in Poly FM. *Delay uses MIDI clk* (menu) locks delay times to the host tempo.

EFFECTS AND AUDIO INPUT

EX Granulator, **EX Multi-FX**, **EX Resonator** and **EX Spectral Freeze** process the plug-in's audio input. Multi-FX and Spectral Freeze are ordinary insert effects. Granulator and Resonator are effects that are also played from MIDI (Audio Unit type *MIDI-controlled effect*): put them where they receive both audio and notes - in Logic they are in the instrument slot's *AU MIDI-controlled Effects* menu with the audio chosen as the side chain; in other hosts insert them on an audio track and route a MIDI track to them. In the standalone apps choose the input under *Options -> Audio/MIDI settings*.

OUTPUTS

Out 1/2 is the main stereo output. **Out 3/4** is a second stereo bus (enable it in your DAW) carrying per-timbre or per-voice outputs depending on the *Output mode*, the paraphonic gate and mono mix (as on the module), and the auto-sampler's CV/gate.

SAVING AND REMEMBERING

- In a DAW, each instance is saved with the project.
- **SAVE** writes a preset to *SD card/presets/<plug-in>*; **LOAD** lists them (plus *Reset to defaults*).
- Every plug-in and app also remembers its **last state**: it is written whenever you change something and when the plug-in closes, and a new instance starts from it. The files live in ~/Library/Application Support/EX-GF (macOS), %APPDATA%\EX-GF (Windows) or ~/.config/EX-GF (Linux).

THE SD CARD FOLDER

The plug-ins read a folder laid out exactly like the disting EX's MicroSD card (default **Music/EX SD Card**).

Folder	Contents
<any name>/	A **multisample folder** of WAV files (see naming below). Every top-level folder that contains WAVs is listed.
wavetables/	Each wavetable is a sub-folder of single-cycle WAVs (used alphabetically), or one WAV of concatenated cycles (600 samples each; 256 for WaveEdit banks - detected automatically). Optional *playlist-wavetable.txt* with -wavelength=.
exciters/	Each exciter table is a folder of up to 128 short WAVs, or one WAV containing exactly 128 equal-length exciters.
FMSYX/	DX7 32-voice bank SysEx files (.syx, 4104 bytes). Without any, three built-in banks are used. MACRO2_1.SYX - MACRO2_3.SYX are the banks of Macro Oscillator 2.
scl/ kbm/	Scala scales and keyboard maps for the Scala/MTS and Scala KBM parameters.
!RECORD/	WAV Recorder recordings (REC00001.WAV …).
!GRANUL8/	Granulator buffers saved with SAVE (GRAN0001.WAV …).
AUTO0001/ …	Auto-sampler output - ordinary multisample folders.
presets/	Presets saved with SAVE.

NAMING MULTISAMPLES

Token	Example	Meaning
_<note>	piano_A2.wav, piano_C#5.wav	Natural pitch. Sharps only, C4 = MIDI note 60.
_SW<n>	piano_A2_SW40.wav	Switch point: the lowest MIDI note that uses this sample.
_V<n>	snare_V1.wav, snare_V2.wav	Velocity layer; V1 is the softest.
_RR<n>	snare_RR1.wav, snare_RR2.wav	Round-robin variants of the same sample.
combined	cello_C3_V2_RR1.wav	Velocity first, then round robin.

- Files without a note name are mapped chromatically from note 48 (C3) upwards - handy for one-shots.
- Without explicit switch points, gaps of up to 3 semitones are covered by pitching the upper sample down; for larger gaps the lower sample is also pitched up over part of the gap (exactly as on the module).
- Loops are read from the WAV's *smpl* or *cue* chunks. A sample loops if it has loop points, otherwise it plays once.
- An optional *playlist.txt* ("disting playlist v1") can set -natural=, -switch=, -loop= and -endLoopOnRelease= per file.

EX MULTISAMPLE



3 - SD MULTISAMPLE

A 16-voice, 3-timbre sample instrument. Choose a folder in the left list; the right list shows its samples - click one to hear it. The display's top line shows the file, natural note and switch point of the highest playing voice.

- **Timbres**: the TIMBRE buttons pick which timbre the folder list sets. Timbres 2 and 3 can follow timbre 1 or use their own folders, and each has its own note range and arpeggiator.
- **SOUND**: gain, saturation, voice count, round-robin mode, output mode and stereo spread.
- **ENV**: attack/decay/sustain/release; release 127 = infinite (the sample plays to its end or loops forever).
- **TUNE**: octave, transpose, fine tune, bend range, Scala tuning and per-voice detune (an "old polysynth" spread).
- **CHORD**: chord generator (key, scale, shape, inversion), arpeggiator per timbre and broken chords (strum time/direction).
- **FX**: stereo or ping-pong delay and bass/treble EQ. **KEYS**: sustain behaviour, velocity curve and note ranges.
- MENU -> SD Multisample -> Options: MIDI note range, paraphonic gate mode, *Delay uses MIDI clk*, *Transpose limit*.

CHORDS AND ARPEGGIOS

Scales	Shapes	Arpeggio modes
Major, Natural Minor, Dominant, Fully Diminished, Dominant Dim, Augmented, Whole Tone, Chromatic, Harmonic Minor	None, Octave, Two Octaves, Root/Fifth, Root/Fifth + 8ve, Triad, Triad + 8ve, Sus4, Sus4 + 8ve, Sixth, Sixth + 8ve, Seventh, Seventh + 8ve, Ninth	Off, Up, Down, Alt, Alt2, Up/Down/Alt/Alt2 -8ve (skip the repeated octave), Step Up, Step Down, Random, Random2 (no repeats), Random3 (permutations), As Played

Each new note (gate) advances the arpeggio by one step through the chord built on it; with chords off, the arpeggio runs over the notes you are holding. *Range* 2 or 3 adds copies one or two octaves up.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Folder	0 .. 999	0	list	
8	Attack time	0 .. 127	0	ENV	
9	Decay time	0 .. 127	60	ENV	
10	Sustain level	0 .. 127	127	ENV	
11	Release time	0 .. 127	77	ENV	
12	Octave	-8 .. 8	0	TUNE	
13	Transpose	-60 .. 60	0	TUNE	
14	Fine tune	-100 .. 100 cents	0 cents	TUNE	
15	Gain	-40 .. 24 dB	0 dB	SOUND	
16	Saturation	0 .. 1	On	SOUND	0=Off, 1=On
17	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
18	Max voices	1 .. 16	16	SOUND	
19	Bend range	0 .. 48	2	TUNE	
20	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
21	Voice 1 detune	-100 .. 100 cents	0 cents	TUNE	
22	Voice 2 detune	-100 .. 100 cents	0 cents	TUNE	
23	Voice 3 detune	-100 .. 100 cents	0 cents	TUNE	
24	Voice 4 detune	-100 .. 100 cents	0 cents	TUNE	
25	Voice 5 detune	-100 .. 100 cents	0 cents	TUNE	
26	Voice 6 detune	-100 .. 100 cents	0 cents	TUNE	
27	Voice 7 detune	-100 .. 100 cents	0 cents	TUNE	
28	Voice 8 detune	-100 .. 100 cents	0 cents	TUNE	
29	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
30	Chord key	-12 .. 12	0	CHORD	
31	Chord scale	0 .. 8	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic, 8=Harmonic Minor
32	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
33	Chord inversion	0 .. 3	0	CHORD	
34	Arpeggio 1 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
35	Arpeggio 2 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played

#	Parameter	Range	Default	Page	Values / notes
36	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
37	Arpeggio 1 range	1 .. 3	1	CHORD	
38	Arpeggio 2 range	1 .. 3	1	CHORD	
39	Arpeggio 3 range	1 .. 3	1	CHORD	
40	Scala/MTS	-16 .. 99	0	TUNE	
41	Scala KBM	0 .. 99	0	TUNE	
42	Folder 2	-1 .. 999	-1	list	
43	Folder 3	-1 .. 999	-1	list	
44	Min note 1	0 .. 127	0	KEYS	
45	Max note 1	0 .. 127	127	KEYS	
46	Min note 2	0 .. 127	0	KEYS	
47	Max note 2	0 .. 127	127	KEYS	
48	Min note 3	0 .. 127	0	KEYS	
49	Max note 3	0 .. 127	127	KEYS	
50	Output spread	-100 .. 100 %	0 %	SOUND	
51	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
52	Delay level	-40 .. 0 dB	-3 dB	FX	
53	Delay time	1 .. 1365 ms	500 ms	FX	
54	Delay feedback	0 .. 100 %	50 %	FX	
55	Tone bass	-240 .. 240	0	FX	
56	Tone treble	-240 .. 240	0	FX	
57	Break time	0 .. 1000 ms	0 ms	CHORD	
58	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate
59	Voice 1 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
60	Voice 2 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
61	Voice 3 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
62	Voice 4 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
63	Voice 5 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
64	Voice 6 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
65	Voice 7 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
66	Voice 8 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Output mode	0 .. 4	All summed 1	SOUND	0=All summed 1, 1=All summed 2, 2=Per gate 12-34, 3=Per gate 12-34 2, 4=Per gate 1-2-3
68	Input mode	0 .. 9	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
69	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
70	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
71	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
72	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
73	Round robin mode	0 .. 3	Sequential	SOUND	0=Sequential, 1=Random, 2=Random2, 3=Random3
74	Voice 9 detune	-100 .. 100 cents	0 cents	TUNE	

#	Parameter	Range	Default	Page	Values / notes
75	Voice 10 detune	-100 .. 100 cents	0 cents	TUNE	
76	Voice 11 detune	-100 .. 100 cents	0 cents	TUNE	
77	Voice 12 detune	-100 .. 100 cents	0 cents	TUNE	
78	Voice 13 detune	-100 .. 100 cents	0 cents	TUNE	
79	Voice 14 detune	-100 .. 100 cents	0 cents	TUNE	
80	Voice 15 detune	-100 .. 100 cents	0 cents	TUNE	
81	Voice 16 detune	-100 .. 100 cents	0 cents	TUNE	



5 - WAV RECORDER AND AUTO-SAMPLER

Switch to the recorder with the **5 WAV RECORDER** tab. **RECORD** records the plug-in's audio input to *!RECORD/RECnnnnn.WAV* (48 or 96 kHz, 16 or 24-bit, mono or stereo); **PLAY** plays the recording selected in the left list. The display shows level meters with peak hold and clip indicators.

AUTO-SAMPLING AN INSTRUMENT

1. **Connect it.** In the standalone app open *Options -> Audio/MIDI settings*: pick the MIDI output that plays your synth and the audio input it comes back on. (In a DAW, route the plug-in's MIDI out to the synth and the synth's audio into the plug-in - not every DAW allows MIDI out of an instrument, so the app is easiest.) Modular synths can use the CV/gate on Out 3/4.
2. **Plan it** on the AUTO-SAMPLER page: start and end note, note step, number of velocity layers (spread between AS Min and AS Max velocity), round robins, note length (AS Coarse seconds + AS Fine ms) and AS Gap (time left for the release).
3. **Press TEST.** A note plays and the status line shows the measured latency in samples. Set *AS Latency* a little below it so each sample starts right at the attack.
4. **Press AUTO.** Every note × velocity × round robin is played and recorded into a new *AUTOnnnn* folder, already named for SD Multisample (e.g. 060_C4_V1_RR1.wav). Press AUTO again to stop early.
5. **Play it.** Click the folder in the right-hand *Auto-sampled* list and the plug-in switches to SD Multisample with it loaded.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Input gain	0 .. 24 dB	0 dB	RECORD	
8	Sample rate	0 .. 1	48kHz	RECORD	0=48kHz, 1=96kHz
9	Bit depth	0 .. 1	16 bit	RECORD	0=16 bit, 1=24 bit
10	Channels	1 .. 6	2	RECORD	
11	Rec CV Mode	0 .. 1	Trigger	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
12	Play CV Mode	0 .. 1	Trigger	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
13	Play As Loop	0 .. 1	One-shot	RECORD	0=One-shot, 1=Loop
14	Outputs 1/2	0 .. 2	Thru/Play	RECORD	0=Thru, 1=Thru/Play, 2=Play
15	Outputs 3/4	0 .. 2	Thru	RECORD	0=Thru, 1=Thru/Play, 2=Play
16	AS Enable	0 .. 1	Off	AUTO-SAMPLER	0=Off, 1=On
17	AS Start Note	0 .. 127	21	AUTO-SAMPLER	
18	AS End Note	0 .. 127	108	AUTO-SAMPLER	
19	AS Note Step	1 .. 127	1	AUTO-SAMPLER	
20	AS Vel Sw	1 .. 9	1	AUTO-SAMPLER	
21	AS RRobins	1 .. 9	1	AUTO-SAMPLER	
22	AS Coarse	0 .. 60 s	1 s	AUTO-SAMPLER	
23	AS Fine	-1000 .. 1000 ms	0 ms	AUTO-SAMPLER	
24	AS Gap	1 .. 600	1	AUTO-SAMPLER	
25	AS Latency	0 .. 960	0	AUTO-SAMPLER	
26	Thru routing 1	0 .. 6	1	RECORD	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
27	Thru routing 2	0 .. 6	2	RECORD	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
28	Thru routing 3	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
29	Thru routing 4	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
30	Thru routing 5	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
31	Thru routing 6	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
32	AS Min velocity	0 .. 127	25	AUTO-SAMPLER	
33	AS Max velocity	0 .. 127	127	AUTO-SAMPLER	
34	Playback file	0 .. 9999	0	list	
35	Record enable	0 .. 1	On	display	0=Locked, 1=On

#	Parameter	Range	Default	Page	Values / notes
36	Playback enable	0 .. 1	On	display	0=Locked, 1=On
37	Playback gain	-40 .. 6 dB	0 dB	RECORD	
38	AS Preview note	0 .. 127	48	AUTO-SAMPLER	

EX POLY WAVETABLE



A 16-voice wavetable synthesizer. Pick a wavetable in the left list; the right list shows its waves - click one to jump the *Wave offset* to it. The position can also follow velocity, pitch, both envelopes and the LFO, and *Wave spread* gives every voice a different position. Each voice has two envelopes (with curve shapes; envelope 2's sustain can be negative), a state-variable filter (low/band/high pass) and an LFO; unison stacks up to 8 of the 16 voices per note.

- **OSC** wave offset/spread, unison, voices, gain · **ENV 1**, **ENV 2** · **FILTER** type, cutoff (as a MIDI note), resonance and modulation · **MOD** routings and LFO speed/retrigger/spread · **CHORD** · **FX** Juno-style chorus I/II, delay, output mode (*Per voice* sends voices to Out 1-4) · **KEYS** tuning, sustain, trigger mode (gated, drone, triggered), Scala.
- Wavetables are band-limited (mip-mapped) so high notes stay clean.

YOUR OWN WAVETABLES AND WAVEEDIT

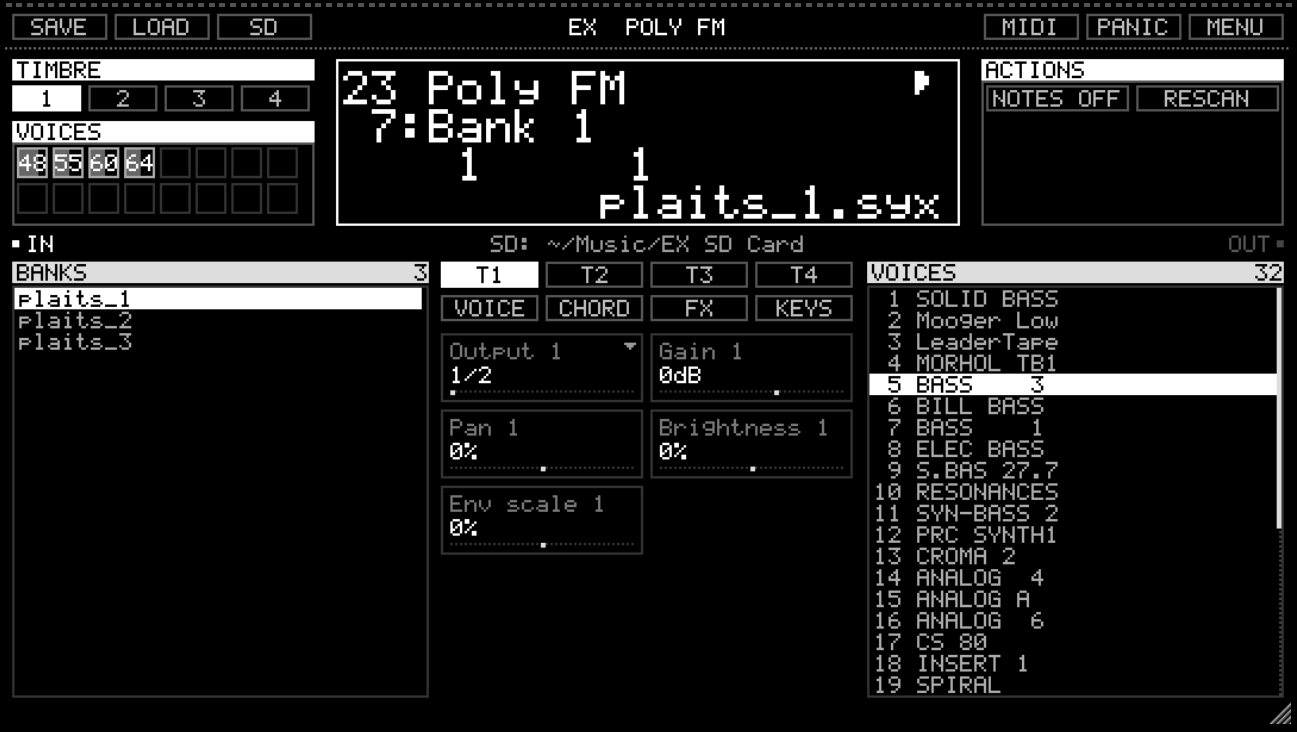
- **Drag and drop** WAV files or folders onto the window, or press **IMPORT**.
- **WAVEEDIT** imports every bank saved by the WaveEdit app in *Documents/WaveEdit/banks*; press it again after editing to update them.
- WaveEdit banks (64 waves × 256 samples) and disting tables (600-sample cycles) are recognised automatically; a set of single-cycle files becomes one table. Each import becomes a folder of single-cycle WAVs in *wavetables/*, so it also works on a real disting EX.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Wavetable	0 .. 999	0	list	
8	Wave offset	-100 .. 100	0	OSC	
9	Wave spread	-100 .. 100	0	OSC	
10	Coarse tune	-60 .. 60	0	KEYS	
11	Fine tune	-100 .. 100 cents	0 cents	KEYS	
12	Attack time	0 .. 127	20	ENV 1	
13	Decay time	0 .. 127	60	ENV 1	
14	Sustain level	0 .. 127	80	ENV 1	
15	Release time	0 .. 127	60	ENV 1	
16	Attack shape	0 .. 127	64	ENV 1	
17	Decay shape	0 .. 127	64	ENV 1	
18	Attack time 2	0 .. 127	60	ENV 2	
19	Decay time 2	0 .. 127	70	ENV 2	
20	Sustain level 2	-127 .. 127	64	ENV 2	
21	Release time 2	0 .. 127	50	ENV 2	
22	Attack shape 2	0 .. 127	64	ENV 2	
23	Decay shape 2	0 .. 127	64	ENV 2	
24	Filter type	0 .. 3	Off	FILTER	0=Off, 1=Lowpass, 2=Bandpass, 3=Highpass
25	Filter freq	0 .. 127	64	FILTER	
26	Filter Q	0 .. 100	50	FILTER	
27	Veloc -> volume	0 .. 100 %	100 %	MOD	
28	Veloc -> wave	-100 .. 100 %	0 %	MOD	
29	Veloc -> filter	-127 .. 127	0	FILTER	
30	Pitch -> wave	-100 .. 100 %	0 %	MOD	
31	Pitch -> filter	-100 .. 100 %	0 %	FILTER	
32	Env -> wave	-100 .. 100 %	0 %	MOD	
33	Env -> filter	-127 .. 127	0	FILTER	
34	Env 2 -> wave	-100 .. 100 %	0 %	MOD	
35	Env 2 -> filter	-127 .. 127	0	FILTER	
36	Env 2 -> pitch	-120 .. 120	0	MOD	
37	LFO -> wave	-100 .. 100 %	0 %	MOD	
38	LFO -> filter	-127 .. 127	0	FILTER	
39	LFO -> pitch	-120 .. 120	0	MOD	
40	LFO speed	-100 .. 100	90	MOD	
41	Gain	-40 .. 24 dB	0 dB	OSC	
42	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
43	Bend range	0 .. 48	2	KEYS	

#	Parameter	Range	Default	Page	Values / notes
44	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
45	Chord key	-12 .. 12	0	CHORD	
46	Chord scale	0 .. 7	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic
47	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
48	Chord inversion	0 .. 3	0	CHORD	
49	Arpeggio 1 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
50	Arpeggio 2 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
51	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
52	Arpeggio 1 range	1 .. 3	1	display	
53	Arpeggio 2 range	1 .. 3	1	display	
54	Arpeggio 3 range	1 .. 3	1	CHORD	
55	Scala/MTS	-16 .. 99	0	KEYS	
56	Scala KBM	0 .. 99	0	KEYS	
57	Chorus mode	0 .. 2	Off	FX	0=Off, 1=I, 2=II
58	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
59	Delay level	-40 .. 0 dB	-3 dB	FX	
60	Delay time	1 .. 10000 ms	500 ms	FX	
61	Delay feedback	0 .. 100 %	50 %	FX	
62	Unison	1 .. 8	1	OSC	
63	Unison detune	0 .. 100 cents	10 cents	OSC	
64	Output spread	-100 .. 100 %	0 %	FX	
65	Output mode	0 .. 3	Spread voice	FX	0=Spread voice, 1=Spread voice 2, 2=Spread pitch, 3=Per voice
66	Input mode	0 .. 18	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
68	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
69	Wave input	0 .. 6	5	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
70	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
71	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
72	Max voices	1 .. 16	16	OSC	
73	LFO retrigger	0 .. 2	Poly	MOD	0=Poly, 1=Mono, 2=Off
74	LFO spread	0 .. 90 deg	0 deg	MOD	
75	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
76	Trigger mode	0 .. 2	Gated	KEYS	0=Gated, 1=Drone, 2=Triggered
77	Break time	0 .. 1000 ms	0 ms	CHORD	
78	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate

EX POLY FM



A 16-voice, 4-timbre FM synthesizer using the six-operator engine of Mutable Instruments Plaits - the same engine the disting EX uses. Banks (left list) are DX7 32-voice SysEx files from the card's *FMSYX* folder; the right list shows the bank's voices by name. The TIMBRE buttons choose which timbre the lists set; timbres 2-4 can follow timbre 1.

- **T1 … T4**: output, gain, pan, *Brightness* (modulator depth) and *Env scale* (envelope times) per timbre.
- **VOICE**: max voices, unison, master gain, velocity -> brightness, voice transpose and output mode. **CHORD**, **FX** (chorus, delay), **KEYS**.
- Parameters 54-64 are not listed in the module's manual; EX-GF uses them for the voice and effect extras.

EDITING WITH A DX7 EDITOR

Point a DX7 editor/librarian (Dexed, Patch Base, MIDI Quest …) at the plug-in. Single-voice dumps, parameter edits and dump requests apply to the timbre chosen by *SysEx target*; the display then shows **VOICE EDITED**. From MENU -> Poly FM: *Write voice* stores the edit in the bank (in memory), *Write bank to SD* saves it, *Abandon edit* reverts, and *Send* dumps the voice or bank back to the editor. *Write built-in to SD* creates FMSYX with the built-in banks.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Bank 1	1 .. 999	1	list	
8	Voice 1	1 .. 32	1	list	
9	Output 1	0 .. 7	1/2	T1	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
10	Gain 1	-40 .. 24 dB	0 dB	T1	
11	Pan 1	-100 .. 100 %	0 %	T1	
12	Brightness 1	-100 .. 100 %	0 %	T1	
13	Env scale 1	-100 .. 100 %	0 %	T1	
14	Bank 2	0 .. 999	0	list	
15	Voice 2	0 .. 32	0	list	
16	Output 2	0 .. 7	1/2	T2	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
17	Gain 2	-40 .. 24 dB	0 dB	T2	
18	Pan 2	-100 .. 100 %	0 %	T2	
19	Brightness 2	-100 .. 100 %	0 %	T2	
20	Env scale 2	-100 .. 100 %	0 %	T2	
21	Bank 3	0 .. 999	0	list	
22	Voice 3	0 .. 32	0	list	
23	Output 3	0 .. 7	1/2	T3	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
24	Gain 3	-40 .. 24 dB	0 dB	T3	
25	Pan 3	-100 .. 100 %	0 %	T3	
26	Brightness 3	-100 .. 100 %	0 %	T3	
27	Env scale 3	-100 .. 100 %	0 %	T3	
28	Bank 4	0 .. 999	0	list	
29	Voice 4	0 .. 32	0	list	
30	Output 4	0 .. 7	1/2	T4	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
31	Gain 4	-40 .. 24 dB	0 dB	T4	
32	Pan 4	-100 .. 100 %	0 %	T4	
33	Brightness 4	-100 .. 100 %	0 %	T4	
34	Env scale 4	-100 .. 100 %	0 %	T4	
35	Coarse tune	-60 .. 60 ST	0 ST	KEYS	
36	Fine tune	-100 .. 100 cents	0 cents	KEYS	
37	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
38	Bend range	0 .. 48	2	KEYS	
39	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
40	Chord key	-12 .. 12	0	CHORD	

#	Parameter	Range	Default	Page	Values / notes
41	Chord scale	0 .. 7	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic
42	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
43	Chord inversion	0 .. 3	0	CHORD	
44	Arpeggio 1 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
45	Arpeggio 2 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
46	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
47	Arpeggio 1 range	1 .. 3	1	display	
48	Arpeggio 2 range	1 .. 3	1	display	
49	Arpeggio 3 range	1 .. 3	1	CHORD	
50	Scala/MTS	-16 .. 99	0	KEYS	
51	Scala KBM	0 .. 99	0	KEYS	
52	Output mode	0 .. 2	Spread voice	VOICE	0=Spread voice, 1=Spread voice 2, 2=Spread pitch
53	Output spread	-100 .. 100 %	0 %	VOICE	
54	Max voices	1 .. 16	16	VOICE	
55	Unison	1 .. 8	1	VOICE	
56	Unison detune	0 .. 100 cents	10 cents	VOICE	
57	Chorus mode	0 .. 2	Off	FX	0=Off, 1=I, 2=II
58	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
59	Delay level	-40 .. 0 dB	-3 dB	FX	
60	Delay time	1 .. 10000 ms	500 ms	FX	
61	Delay feedback	0 .. 100 %	50 %	FX	
62	Gain	-40 .. 24 dB	0 dB	VOICE	
63	Veloc -> bright	-100 .. 100 %	0 %	VOICE	
64	Voice transpose	0 .. 1	Use	VOICE	0=Ignore, 1=Use
65	Input mode	0 .. 9	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
66	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
67	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
68	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
69	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
70	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
71	Trigger mode	0 .. 2	Gated	KEYS	0=Gated, 1=Drone, 2=Triggered
72	Break time	0 .. 1000 ms	0 ms	CHORD	
73	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate
74	SysEx target	1 .. 4	1	KEYS	

EX POLY EXCITER



A 16-voice synthesizer that excites a resonating string with a short burst of sound - plucked, struck, scraped or bowed. The left list holds exciter tables (*Internal*, or tables from the card's *exciters* folder); the right list shows the exciters in the table. Drum hits make great exciters: a kick for a soft tone, a snare for a sharp, noisy attack.

Mode	Behaviour
Strike	The exciter plays once, for the whole Exciter length.
Scrape	As Strike, but the exciter stops when the key is released.
Strum	The exciter loops while the key is held; its level follows **pressure** (aftertouch or the Pressure parameter) - silent without it.

Internal exciter	Sound
0 Ramp	A decaying linear ramp.
1 Noise burst	White noise at a constant level.
2 Noise ramp	Noise shaped by a decaying ramp.
3 Square	Constant level: strong plucks at the start and end.
4 Silence	For processing external audio only.

- **EXCITER** mode, exciter, length, pressure, *Ext exciter input* (feed audio into the plug-in to excite the strings live).
- **RESONATOR** tone (brightness), damping, extra damping on release, low-pass filter. **VELO** velocity/pitch/pressure modulation. **ENV/LFO** modulate pitch and exciter choice. **CHORD**, **FX**, **KEYS**.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Mode	0 .. 2	Strike	EXCITER	0=Strike, 1=Scrape, 2=Strum
8	Table	-1 .. 999	-1	list	
9	Exciter	0 .. 127	0	EXCITER	
10	Exciter length	1 .. 597 ms	100 ms	EXCITER	
11	Tone	0 .. 127	50	RESONATOR	
12	Damp	0 .. 100	4	RESONATOR	
13	Release damp	0 .. 100	0	RESONATOR	
14	Pressure	0 .. 127	0	EXCITER	
15	Veloc -> strike	0 .. 100 %	100 %	VELO	
16	Veloc -> tone	-100 .. 100 %	0 %	VELO	
17	Veloc -> damp	-100 .. 100 %	0 %	VELO	
18	Veloc -> exciter	-127 .. 127	0	VELO	
19	Pitch -> damp	-100 .. 100 %	0 %	VELO	
20	Pitch -> exciter	-100 .. 100 %	0 %	VELO	
21	Pitch -> LPF	-100 .. 100 %	0 %	RESONATOR	
22	Press -> exciter	-127 .. 127	0	VELO	
23	Env -> pitch	-120 .. 120	0	ENV/LFO	
24	Env -> exciter	-127 .. 127	0	ENV/LFO	
25	LFO -> pitch	-200 .. 200 cents	0 cents	ENV/LFO	
26	LFO -> exciter	-127 .. 127	0	ENV/LFO	
27	LFO speed	-100 .. 100	90	ENV/LFO	
28	Attack time	0 .. 127	60	ENV/LFO	
29	Decay time	0 .. 127	70	ENV/LFO	
30	Sustain level	-127 .. 127	0	ENV/LFO	
31	Release time	0 .. 127	50	ENV/LFO	
32	Coarse tune	-60 .. 60	0	KEYS	
33	Fine tune	-100 .. 100 cents	0 cents	KEYS	
34	Gain	-40 .. 24 dB	0 dB	RESONATOR	
35	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
36	Bend range	0 .. 48	2	KEYS	
37	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
38	Chord key	-12 .. 12	0	CHORD	
39	Chord scale	0 .. 7	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic

#	Parameter	Range	Default	Page	Values / notes
40	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
41	Chord inversion	0 .. 3	0	CHORD	
42	Arpeggio 1 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
43	Arpeggio 2 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
44	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
45	Arpeggio 1 range	1 .. 3	1	display	
46	Arpeggio 2 range	1 .. 3	1	display	
47	Arpeggio 3 range	1 .. 3	1	CHORD	
48	Scala/MTS	-16 .. 99	0	KEYS	
49	Scala KBM	0 .. 99	0	KEYS	
50	Low pass filter	0 .. 127	126	RESONATOR	
51	Chorus mode	0 .. 2	Off	FX	0=Off, 1=I, 2=II
52	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
53	Delay level	-40 .. 0 dB	-3 dB	FX	
54	Delay time	1 .. 10000 ms	500 ms	FX	
55	Delay feedback	0 .. 100 %	50 %	FX	
56	Unison	1 .. 8	1	FX	
57	Unison detune	0 .. 100 cents	10 cents	FX	
58	Output spread	-100 .. 100 %	0 %	FX	
59	Break time	0 .. 1000 ms	0 ms	CHORD	
60	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate
61	Output mode	0 .. 3	Spread voice	FX	0=Spread voice, 1=Spread voice 2, 2=Spread pitch, 3=Per voice
62	Input mode	0 .. 18	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
63	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
64	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
65	Ext exciter input	0 .. 6	None	EXCITER	0=None, 1=In L, 2=In R, 3=In L+R, 4=In L+R, 5=In L+R, 6=In L+R
66	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
68	Max voices	1 .. 16	16	KEYS	
69	LFO retrigger	0 .. 1	Poly	ENV/LFO	0=Poly, 1=Off
70	LFO spread	0 .. 90 deg	0 deg	ENV/LFO	
71	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*

EX GRANULATOR



Granular synthesis: many short snippets (grains) of the audio in a buffer, played with randomised timing, length, pitch and pan. The buffer (mono, up to 31.25 s) is recorded from the audio input or loaded from a WAV file. Grain clouds are started by **notes**: MIDI notes, or the three **drones** you simply switch on. Nothing sounds until the buffer holds audio.

- **Two ways to start:** feed audio in, press **RECORD** and switch on **DRONE 1**; or load a WAV and play notes.
- **Lists:** the left list shows the folders on the SD card that contain WAVs (plus !GRANUL8 and !RECORD), the right list the WAV files in that folder. Click a file to load it, or drag a WAV onto the window. The buffer size is set to the file's length.
- **BUFFER**: size, input gain, record, record fade. *Input feedback* echoes the input at the buffer length; *Main feedback* records the grains back into the buffer. The write position only moves while recording.
- **GRAINS**: spawn mode, rate, size, window shape, opacity (the share of grains that actually sound), reverse probability, grain limit and normalize. *Size mean* 0 makes each grain one cycle of the note's pitch.
- **PITCH**: pitch mean and spread (*Pitch quantize* snaps the random part to octaves, fourths, fifths or triads) and pan. *Natural pitch* is the note that plays the audio at its recorded pitch.
- **POS**: *Delay mean/spread* set where in the buffer grains start, as a percentage of the buffer or in ms (*Delay unit*). The LFO sweeps that position; at speed 196 and depth 100% it moves at normal playback speed.
- **MOD**: note envelope (100 ms - 30 s), and how envelope, velocity and pitch affect opacity, level, pitch and position. **DRONES**: pitch, opacity and level of the three drones.
- **FX** and **REVERB**: stereo or ping-pong delay and a reverb. *Effects routing* decides whether they act on the grains only or on the mix with the dry input.
- **VISUALS** shows the buffer waveform with the grain positions, the number of grains, the played range, and the delay mean and buffer size in ms. **SAVE** writes the buffer to *!GRANUL8/GRANnnnn.WAV*. A loaded or saved file is remembered and reloaded with the preset or project.
- Out 1/2 is the mix with effects; Out 3/4 is the grains alone.

Spawn mode	Behaviour
Stochastic	Grains start at random intervals set by Rate mean and Rate spread.
Mid-grain	A new grain starts at the middle of the current one. With Size spread 0 and the Equal power shape this loops smoothly.

Spawn mode	Behaviour
… (fixed)	As above, but grain positions are measured from the write position at the moment the note started, so a note keeps its timbre while recording continues. Without recording there is no difference.
Single	Exactly one grain per note.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Buffer size	100 .. 31250 ms	5000 ms	BUFFER	
8	Input gain	-34 .. 12 dB	0 dB	BUFFER	
9	Dry gain	-40 .. 6 dB	0 dB	FX	
10	Granulator gain	-40 .. 6 dB	0 dB	FX	
11	Record	0 .. 1	Off	BUFFER	0=Off, 1=On
12	Spawn mode	0 .. 4	Stochastic	GRAINS	0=Stochastic, 1=Mid-grain, 2=Stochastic (fixed), 3=Mid-grain (fixed), 4=Single
13	Rate mean	1 .. 1000 ms	5 ms	GRAINS	
14	Rate spread	0 .. 200 %	10 %	GRAINS	
15	Size mean	0 .. 1000 ms	100 ms	GRAINS	
16	Size spread	0 .. 200 %	10 %	GRAINS	
17	Pitch mean	-24 .. 24 ST	0 ST	PITCH	
18	Pitch spread	0 .. 1200 cents	0 cents	PITCH	
19	Pan mean	-100 .. 100 %	0 %	PITCH	
20	Pan spread	0 .. 100 %	10 %	PITCH	
21	Delay mean	0 .. 5000	2500	POS	
22	Delay spread	0 .. 5000	250	POS	
23	Shape	0 .. 5	Gaussian	GRAINS	0=Gaussian, 1=Tukey, 2=Triangle, 3=Expodec, 4=Rexpodec, 5=Equal power
24	Opacity	0 .. 100 %	100 %	GRAINS	
25	LFO depth	-100 .. 100 %	0 %	POS	
26	LFO speed	0 .. 255	196	POS	
27	Input feedback	0 .. 100 %	0 %	BUFFER	
28	Main feedback	0 .. 100 %	0 %	BUFFER	
29	Attack time	0 .. 127	64	MOD	
30	Release time	0 .. 127	64	MOD	
31	Env -> opacity	0 .. 100 %	0 %	MOD	
32	Env -> level	0 .. 100 %	100 %	MOD	
33	Veloc -> level	0 .. 100 %	100 %	MOD	
34	Veloc -> delay	-100 .. 100 %	0 %	MOD	
35	Pitch -> pitch	-100 .. 100 %	100 %	MOD	
36	Pitch -> delay	-100 .. 100 %	0 %	MOD	
37	Normalize	0 .. 1	On	GRAINS	0=Off, 1=On
38	Natural pitch	0 .. 127 ST	48 ST	PITCH	
39	Drone 1 pitch	0 .. 127 ST	48 ST	DRONES	
40	Drone 2 pitch	0 .. 127 ST	36 ST	DRONES	
41	Drone 3 pitch	0 .. 127 ST	60 ST	DRONES	
42	Drone 1 enable	0 .. 1	Off	DRONES	0=Off, 1=On
43	Drone 2 enable	0 .. 1	Off	DRONES	0=Off, 1=On

#	Parameter	Range	Default	Page	Values / notes
44	Drone 3 enable	0 .. 1	Off	DRONES	0=Off, 1=On
45	Drone 1 opacity	0 .. 100 %	100 %	DRONES	
46	Drone 2 opacity	0 .. 100 %	100 %	DRONES	
47	Drone 3 opacity	0 .. 100 %	100 %	DRONES	
48	Drone 1 level	-40 .. 6 dB	0 dB	DRONES	
49	Drone 2 level	-40 .. 6 dB	0 dB	DRONES	
50	Drone 3 level	-40 .. 6 dB	0 dB	DRONES	
51	Out delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
52	Out delay level	-40 .. 0 dB	-3 dB	FX	
53	Out delay time	1 .. 2730 ms	500 ms	FX	
54	Out delay feedback	0 .. 100 %	50 %	FX	
55	Reverb mix	0 .. 100 %	50 %	FX	
56	Reverb time	400 .. 30000 ms	1700 ms	FX	
57	Grain limit	1 .. 40	40	GRAINS	
58	Bend range	0 .. 48 ST	2 ST	PITCH	
59	Reverse	0 .. 100 %	0 %	GRAINS	
60	Pitch quantize	0 .. 5	Off	PITCH	0=Off, 1=Octaves, 2=Fourths, 3=Fifths, 4=Major Triad, 5=Minor Triad
61	LFO shape	0 .. 2	Triangle	POS	0=Triangle, 1=Ramp up, 2=Ramp down
62	Record fade	0 .. 1000 ms	5 ms	BUFFER	
63	Reserved 63	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
64	Reserved 64	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
65	Input mode	0 .. 3	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
66	Mono input	0 .. 1	Stereo	BUFFER	0=Stereo, 1=Mono (L)
67	Sustain mode	0 .. 1	Synth	MOD	0=Synth, 1=Piano
68	Delay mean input	0 .. 12	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
69	Effects routing	0 .. 2	Dly>Rev>Mix	FX	0=Dly>Rev>Mix, 1=Dly>Mix>Rev, 2=Mix>Dly>Rev
70	Reverb model	0 .. 3	Hall	REVERB	0=Room, 1=Hall, 2=Plate, 3=Cathedral
71	Reverb size	1 .. 100 %	100 %	REVERB	
72	Reverb high damp	0 .. 100	60	REVERB	
73	Reverb mod speed	1 .. 500	250	REVERB	
74	Reverb mod depth	0 .. 100	25	REVERB	
75	Reverb early gain	-40 .. 6 dB	-12 dB	REVERB	
76	Reverb diff gain	-40 .. 6 dB	-6 dB	REVERB	
77	Delay unit	0 .. 1	%	POS	0=%, 1=ms

EX MULTI-FX



A stereo multi-effects processor: a 3-band EQ feeding a pitch section, a delay and a reverb, all running at once, with free routing between them and a variable sample rate. The left list holds 13 **programs** to start from; choosing one resets every parameter and then applies the program.

- **Routing**: each of the pitch, delay and reverb sections takes a mix of two sources (*input 1*, *input 2*, *input mix*) chosen from Pre-EQ, Post-EQ, Pitch, Delay or Reverb. Sections can be chained in any order, run in parallel, or fed back into each other. Feedback loops can howl - route with care.
- **MIX**: dry gain (pre- or post-EQ), overall effects gain, one gain per section, saturation and mono input. *Aux outputs* chooses which section appears on Out 3/4.
- **EQ**: band 1 is a 6 or 12 dB/octave high-pass or a low shelf; band 2 a peak/notch with Q; band 3 a 6 or 12 dB/octave low-pass or a high shelf. Frequencies are set as MIDI note numbers (69 = 440 Hz).
- **PITCH**: chorus/flanger, phaser or pitch shifter. For a chorus use a longish *Chorus delay* with low depth and feedback; for a flanger a short delay, high depth and feedback to taste. *Spread* 180° gives the classic stereo chorus. In phaser mode *Chorus delay* sets the centre frequency. *Blend* below zero crossfades dry to wet; above zero it adds wet to dry.
- **DELAY**: mono, stereo (separate left and right times) or ping-pong (with *mono-ize* and *initial pan*). Each has a *fade* version, which crossfades to a new time, and a *slew* version, which glides like tape. **TAP** sets the time; MENU -> Multi FX -> Options -> *Delay uses MIDI clk* follows the host tempo.
- **REVERB**: four models, time from 0.4 to 30 s, size, high damping, modulation, early and diffuse levels, input low-pass.
- **VSR** (variable sample rate) lowers the internal sample rate in semitones. Every EQ frequency drops, delay and reverb times stretch, and the tails already in the buffers change pitch. -24 gives a 12 kHz retro sound and four times the delay time. *Quantize* snaps the rate to a scale.
- **BYPASS** passes the input through, **CLEAR** empties the delay and reverb. The LEVELS cells show input, pitch, delay and reverb activity; the display shows CLIP when the output overloads.

The reverb model names (Room, Hall, Plate, Cathedral) and the Quantize scale names are EX-GF's own - the module's manual does not list them. Parameters that select CV inputs are kept for numbering only.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	VSR	-80 .. 80 ST	0 ST	VSR	
8	VSR attenuator	0 .. 100 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
9	Quantize	0 .. 16	Off	VSR	
10	Dry routing	0 .. 1	Pre-EQ	MIX	0=Pre-EQ, 1=Post-EQ
11	Dry gain	-40 .. 6 dB	0 dB	MIX	
12	Effects gain	-40 .. 6 dB	0 dB	MIX	
13	Pitch gain	-40 .. 6 dB	0 dB	MIX	
14	Delay gain	-40 .. 6 dB	0 dB	MIX	
15	Reverb gain	-40 .. 6 dB	0 dB	MIX	
16	Saturation	0 .. 1	On	MIX	0=Off, 1=On
17	EQ 1 type	0 .. 2	Low shelf	EQ	0=HP 6dB, 1=HP 12dB, 2=Low shelf
18	EQ 1 frequency	0 .. 127	26	EQ	
19	EQ 1 gain	-120 .. 120 0.1dB	0 0.1dB	EQ	
20	EQ 2 frequency	0 .. 127	75	EQ	
21	EQ 2 Q	0 .. 127	64	EQ	
22	EQ 2 gain	-120 .. 120 0.1dB	0 0.1dB	EQ	
23	EQ 3 type	0 .. 2	High shelf	EQ	0=LP 6dB, 1=LP 12dB, 2=High shelf
24	EQ 3 frequency	0 .. 127	118	EQ	
25	EQ 3 gain	-120 .. 120 0.1dB	0 0.1dB	EQ	
26	Pitch input 1	0 .. 4	Post-EQ	PITCH	0=Pre-EQ, 1=Post-EQ, 2=Pitch, 3=Delay, 4=Reverb
27	Pitch input 2	0 .. 4	Delay	PITCH	0=Pre-EQ, 1=Post-EQ, 2=Pitch, 3=Delay, 4=Reverb
28	Pitch input mix	-100 .. 100 %	-100 %	PITCH	
29	Pitch effect	0 .. 2	Chorus	PITCH	0=Chorus, 1=Phaser, 2=Pitch shift
30	Pitch blend	-100 .. 100 %	100 %	PITCH	
31	Pitch feedback	-100 .. 100 %	0 %	PITCH	
32	Chorus delay	1 .. 853 0.1ms	580 0.1ms	PITCH	
33	Chorus speed	1 .. 200 0.01Hz	40 0.01Hz	PITCH	
34	Chorus depth	0 .. 100	15	PITCH	
35	Chorus spread	0 .. 360 deg	180 deg	PITCH	
36	Pitch shift delay	1 .. 200 ms	200 ms	PITCH	
37	Pitch shift	-48 .. 48 ST	0 ST	PITCH	
38	Pitch shift (fine)	-100 .. 100 cents	0 cents	PITCH	
39	Delay input 1	0 .. 4	Post-EQ	DELAY	0=Pre-EQ, 1=Post-EQ, 2=Pitch, 3=Delay, 4=Reverb
40	Delay input 2	0 .. 4	Pitch	DELAY	0=Pre-EQ, 1=Post-EQ, 2=Pitch, 3=Delay, 4=Reverb
41	Delay input mix	-100 .. 100 %	-100 %	DELAY	

#	Parameter	Range	Default	Page	Values / notes
42	Delay mode	0 .. 5	Mono fade	DELAY	0=Mono fade, 1=Stereo fade, 2=Ping-pong fade, 3=Mono slew, 4=Stereo slew, 5=Ping-pong slew
43	Delay time	1 .. 5000 ms	500 ms	DELAY	
44	Delay time R	1 .. 5000 ms	500 ms	DELAY	
45	Delay feedback	0 .. 100 %	50 %	DELAY	
46	Delay mono-ize	0 .. 100 %	100 %	DELAY	
47	Delay initial pan	-100 .. 100	-100	DELAY	
48	Reverb input 1	0 .. 4	Post-EQ	REVERB	0=Pre-EQ, 1=Post-EQ, 2=Pitch, 3=Delay, 4=Reverb
49	Reverb input 2	0 .. 4	Delay	REVERB	0=Pre-EQ, 1=Post-EQ, 2=Pitch, 3=Delay, 4=Reverb
50	Reverb input mix	-100 .. 100 %	-100 %	REVERB	
51	Reverb model	0 .. 3	Hall	REVERB	0=Room, 1=Hall, 2=Plate, 3=Cathedral
52	Reverb time	400 .. 30000 ms	1000 ms	REVERB	
53	Reverb size	1 .. 100 %	100 %	REVERB	
54	Reverb high damp	0 .. 100	60	REVERB	
55	Reverb mod speed	1 .. 500 0.01Hz	250 0.01Hz	REVERB	
56	Reverb mod depth	0 .. 100	25	REVERB	
57	Reverb early gain	-40 .. 6 dB	-12 dB	REVERB	
58	Reverb diff gain	-40 .. 6 dB	-6 dB	REVERB	
59	Reverb low pass	0 .. 127	100	REVERB	
60	Mono input	0 .. 1	Off	MIX	0=Off, 1=On
61	VSR input	0 .. 6	Input 3	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
62	Pitch shift input	0 .. 6	Input 4	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
63	Aux outputs	0 .. 4	Pitch	MIX	0=None, 1=Pitch, 2=Delay, 3=Reverb, 4=All effects
64	Aux input L	0 .. 6	None	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
65	Aux input R	0 .. 6	None	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
66	Aux input gain	-40 .. 6 dB	0 dB	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Delay clock input	0 .. 6	None	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
68	Delay R clock input	0 .. 6	None	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
69	Delay multiplier	0 .. 23	15	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
70	Delay R multiplier	0 .. 23	15	-	*CV/hardware only - kept for numbering, no effect in the plug-in*

EX RESONATOR



The disting EX's port of Mutable Instruments **Rings**, using the original DSP code. It resonates the audio input, or plays its own exciter when you send notes. Demos and tutorials for Rings apply to it directly.

Mode	What it is
Modal	A bank of resonant band-pass filters: bars, bells, plates.
Sympathetic strings	A string with sympathetic strings tuned by Structure.
String	A plucked/bowed string with dispersion (inharmonicity).
FM voice	A two-operator FM voice driven by the input's envelope.
Symp. quantized	Sympathetic strings tuned to the chosen chord.
String + reverb	The string model followed by a reverb.
String synth	Rings' hidden string-machine mode, with six effects (formant filter, chorus, reverb, and brighter variants).

- **Lists**: the left list selects the mode; the right list the chord (for the chord modes) or the String synth effect.
- **MODEL**: polyphony (1-4 notes ring at once), *Structure*, *Brightness*, *Damping* and *Position* as on Rings.
- **Playing**: *MIDI mode* chooses what a note does - set the pitch, strum, or both (the default here; the module's default is Off). With MIDI mode Off or Pitch, Rings strums itself from pitch changes or from transients in the input, exactly as the module does with nothing patched to Strum. **STRUM** (or pushing L) strums by hand.
- **INPUT** off (*Audio input* = None) makes Rings use its internal exciter instead of the audio input.
- **MIX**: input gain, noise gate, output gain, dry gain, and pan positions for the odd output, the even output and the dry signal. The dry signal is at 0 dB by default, as on the module; **DRY** mutes it.
- Out 1/2 is the stereo mix, Out 3 is odd + even, Out 4 is a mono mix of the stereo output. The engine runs at 48 kHz internally and is resampled at other host rates.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Mode	0 .. 6	Modal	MODEL	0=Modal, 1=Sympathetic strings, 2=String, 3=FM voice, 4=Symp. quantized, 5=String + reverb, 6=String synth
8	Synth effect	0 .. 5	Formant	MODEL	0=Formant, 1=Chorus, 2=Reverb, 3=Formant 2, 4=Ensemble, 5=Reverb 2
9	Polyphony	1 .. 4	1	MODEL	
10	Coarse tune	-60 .. 60 ST	0 ST	TUNE	
11	Fine tune	-100 .. 100 cents	0 cents	TUNE	
12	Structure	0 .. 127	64	MODEL	
13	Brightness	0 .. 127	64	MODEL	
14	Damping	0 .. 127	64	MODEL	
15	Position	0 .. 127	64	MODEL	
16	Chord	0 .. 10	Octaves	MODEL	0=Octaves, 1=Fifths, 2=Sus 4, 3=Minor, 4=Minor 7, 5=Minor 9, 6=Minor 11, 7=6/9, 8=Major 9, 9=Major 7, 10=Major
17	Noise gate	0 .. 1	On	MIX	0=Off, 1=On
18	Input gain	-40 .. 12 dB	0 dB	MIX	
19	Output gain	-40 .. 12 dB	0 dB	MIX	
20	Dry gain	-40 .. 12 dB	0 dB	MIX	
21	Odd pan	-100 .. 100 %	-100 %	MIX	
22	Even pan	-100 .. 100 %	100 %	MIX	
23	Dry pan	-100 .. 100 %	0 %	MIX	
24	Strum input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
25	V/Oct input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
26	Audio input	0 .. 6	Input	TUNE	0=None, 1=Input, 2=Input, 3=Input, 4=Input, 5=Input, 6=Input
27	Brightness input	0 .. 6	5	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
28	Frequency input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
29	Damping input	0 .. 6	6	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
30	Structure input	0 .. 6	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
31	Position input	0 .. 6	3	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
32	MIDI mode	0 .. 3	Pitch & strum	TUNE	0=Off, 1=Pitch, 2=Strum, 3=Pitch & strum
33	I2C mode	0 .. 3	Off	-	*CV/hardware only - kept for numbering, no effect in the plug-in*

EX SPECTRAL FREEZE



An audio freeze: it captures the sound at one moment and sustains it for as long as you like. It works on the spectrum (a phase vocoder), so the result is a smooth sustained texture rather than a loop, and it can also be played back at any speed and pitch. There are four independent freeze voices.

- **FREEZE** (or pushing L) freezes. With Target **A** (auto), the first press freezes the next free voice, the second releases it and the third freezes a new one - so a double press crossfades to a fresh capture.
- **TARGET 1-4**: FREEZE toggles that voice, so you can capture up to four different moments and play them together. The left **Voices** list shows each voice's state; click a row to freeze or release it.
- **MOVE**: a single frozen frame is completely static, so the voice can move around in the audio just before the freeze point. *Depth* sets how much audio is used (in FFT frames, shown in ms; 1 = static), *Rate* the speed (0 = stopped, 100 = original speed, shown as a slow-down factor) and *Movement* the direction. *Offset* moves the freeze point back in time.
- *Etherization* below 127 progressively drops the transient parts of the sound, leaving a few pure tones.
- **LIVE** (*Live process*) starts a voice that resynthesizes the live input instead of a frozen moment - a way to use etherization and pitch shift on live audio. **UNFREEZE** releases all voices.
- **VOICES** and **PITCH**: gain, pan and coarse/fine pitch shift per voice, added to the global shift. *Attack* and *Release* set the fade times.
- **MIX** and **FX**: dry and freeze gains, saturation, a delay and a reverb. Each effect can sit on the freeze output only or on the whole output mix; the delay comes first.
- **VISUALS** switches the module display to the four-quadrant view: envelope bar, the segment in use and the play position, with a T on the target voice.
- Out 1/2 is the mix; Out 3/4 is the dry input.

Movement	Behaviour
Forwards	Plays the audio forwards in a loop (the loop point is interpolated, not crossfaded).
Backwards	The same in reverse.
Alternating	Forwards and backwards in turn.
Random walk	The position moves continuously but changes direction at random.
Random skip	The position jumps to random places.

MIDI note	Action
C3, D3, E3, F3 (48, 50, 52, 53)	Set the freeze target to voice 1-4.
G3 (55) / A3 (57) / B3 (59)	Freeze toggle / freeze gate (while held) / target Auto.
C4, D4, E4, F4 (60, 62, 64, 65)	Toggle voices 1-4.
C5, D5, E5, F5 (72, 74, 76, 77)	Freeze voices 1-4 while the key is held.

MIDI control works in the VST3 and LV2 versions and the app; Audio Unit hosts do not send MIDI to insert effects.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Freeze toggle	0 .. 1	Off	FREEZE	0=Off, 1=On
8	Freeze gate	0 .. 1	Off	FREEZE	0=Off, 1=On
9	Freeze target	0 .. 4	2	FREEZE	0=Auto, 1=1, 2=2, 3=3, 4=4
10	Voice 1 freeze	0 .. 1	Off	VOICES	0=Off, 1=On
11	Voice 2 freeze	0 .. 1	Off	VOICES	0=Off, 1=On
12	Voice 3 freeze	0 .. 1	Off	VOICES	0=Off, 1=On
13	Voice 4 freeze	0 .. 1	Off	VOICES	0=Off, 1=On
14	Movement	0 .. 4	Alternating	MOVE	0=Forwards, 1=Backwards, 2=Alternating, 3=Random walk, 4=Random skip
15	Rate	0 .. 100	1	MOVE	
16	Depth	1 .. 128	2	MOVE	
17	Offset	0 .. 1023	0	MOVE	
18	Etherization	0 .. 127	127	FREEZE	
19	Attack	0 .. 127	10	FREEZE	
20	Release	0 .. 127	10	FREEZE	
21	Dry gain	-40 .. 6 dB	0 dB	MIX	
22	Freeze gain	-40 .. 6 dB	0 dB	MIX	
23	Voice 1 gain	-40 .. 6 dB	0 dB	VOICES	
24	Voice 2 gain	-40 .. 6 dB	0 dB	VOICES	
25	Voice 3 gain	-40 .. 6 dB	0 dB	VOICES	
26	Voice 4 gain	-40 .. 6 dB	0 dB	VOICES	
27	Voice 1 pan	-100 .. 100 %	0 %	VOICES	
28	Voice 2 pan	-100 .. 100 %	0 %	VOICES	
29	Voice 3 pan	-100 .. 100 %	0 %	VOICES	
30	Voice 4 pan	-100 .. 100 %	0 %	VOICES	
31	Coarse shift	-48 .. 24 ST	0 ST	PITCH	
32	Fine shift	-100 .. 100 cents	0 cents	PITCH	
33	Voice 1 coarse	-48 .. 24 ST	0 ST	PITCH	
34	Voice 2 coarse	-48 .. 24 ST	0 ST	PITCH	
35	Voice 3 coarse	-48 .. 24 ST	0 ST	PITCH	
36	Voice 4 coarse	-48 .. 24 ST	0 ST	PITCH	
37	Voice 1 fine	-100 .. 100 cents	0 cents	PITCH	
38	Voice 2 fine	-100 .. 100 cents	0 cents	PITCH	
39	Voice 3 fine	-100 .. 100 cents	0 cents	PITCH	
40	Voice 4 fine	-100 .. 100 cents	0 cents	PITCH	
41	Out delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
42	Out delay level	-40 .. 0 dB	-3 dB	FX	
43	Out delay time	1 .. 2730 ms	500 ms	FX	

#	Parameter	Range	Default	Page	Values / notes
44	Out delay feedback	0 .. 100 %	50 %	FX	
45	Reverb mix	0 .. 100 %	0 %	FX	
46	Reverb time	400 .. 30000 ms	1700 ms	FX	
47	Reverb model	0 .. 3	Hall	FX	0=Room, 1=Hall, 2=Plate, 3=Cathedral
48	Reverb size	1 .. 100 %	100 %	FX	
49	Reverb high damp	0 .. 100	60	FX	
50	Reverb mod speed	1 .. 500	50	FX	
51	Reverb mod depth	0 .. 100	10	FX	
52	Reverb early gain	-40 .. 6 dB	-12 dB	FX	
53	Reverb diff gain	-40 .. 6 dB	-6 dB	FX	
54	Live process	0 .. 1	Off	FREEZE	0=Off, 1=On
55	Unfreeze all	0 .. 1	Off	FREEZE	0=Off, 1=On
56	Reserved 56	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
57	Reserved 57	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
58	Reserved 58	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
59	Reserved 59	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
60	Reserved 60	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
61	Reserved 61	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
62	Reserved 62	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
63	Reserved 63	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
64	Reserved 64	0 .. 1	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
65	Mono input	0 .. 1	Mono	MIX	0=Stereo, 1=Mono
66	Delay routing	0 .. 1	Freeze output	MIX	0=Freeze output, 1=Output mix
67	Reverb routing	0 .. 1	Freeze output	MIX	0=Freeze output, 1=Output mix
68	Saturation	0 .. 1	Off	MIX	0=Off, 1=On

EX MACRO OSCILLATOR 2



Four instances of Mutable Instruments **Plaits** (firmware 1.2, original DSP code). Out of the box it is a 4-voice synth: the four oscillators take notes in turn (round-robin) and share one set of settings. Demos and tutorials for Plaits apply to it directly.

- **OSCILLATOR 1-4** chooses which oscillator the lists edit. The left list holds the 24 **models**; with one of the 6-op FM models selected, the right list shows the 32 voices of its bank (the *Harmonics* control selects the voice).
- **OSC 1-4** pages: *Harmonics*, *Timbre*, *Morph*, FM and modulation amounts (driven by Plaits' internal decay envelope), *LP gate* colour, *Time/decay* and tuning.
- **Shared settings**: *Share set* (MIDI page) says whose settings an oscillator uses. By default all four use oscillator 1's, so the OSC 2-4 pages have no effect until you press **UNLINK**, which gives every oscillator its own - then each note of a chord can use a different model. (The module's default is unlinked.)
- **MIDI option** per oscillator: *Off* (the oscillator drones), *Pitch*, *Pitch/trig*, *Trig*, *Trig by* one note (for the drum models) or *Poly* (round-robin group), each on the receive channel or one of the next three. The default here is Poly for all four; the module's default is Off.
- **Envelope**: *Trigger* pings the low-pass gate as a trigger does on Plaits, and the sound decays by *Time/decay*. *Gate* (**GATE ENV**) holds the sound while the key is down. Velocity sets the level - or the accent in the drum, string, modal and 6-op FM models. The 6-op FM voices sustain while the key is held in either mode.
- **Outputs**: in *Mixed* mode Out 1 (left) carries all the main outputs and Out 2 (right) all the aux outputs; Out 3/4 repeat them. **SPLIT OUT** (*Individual*) assigns any oscillator's main or aux output to each of Out 1-4.
- **FM banks**: FMSYX/MACRO2_1.SYX, MACRO2_2.SYX and MACRO2_3.SYX on the SD card replace the three built-in banks. MENU -> Macro Oscillator 2 -> *Write FM banks to SD* creates them from the built-in banks.
- The engine runs at 48 kHz internally and is resampled at other host rates.

#	Model	#	Model	#	Model
0	Virtual analog	8	Granular cloud	16	Virtual analog with filter
1	Waveshaper	9	Filtered noise	17	Phase distortion
2	2-op FM	10	Particle noise	18	6-op FM 1
3	Granular formant	11	Inharmonic string	19	6-op FM 2
4	Harmonic	12	Modal resonator	20	6-op FM 3

#	Model	#	Model	#	Model
5	Wavetable	13	Analog bass drum	21	Wave terrain
6	Chords	14	Analog snare	22	String machine
7	Speech	15	Analog hi-hat	23	Chiptune

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Model 1	0 .. 23	0	OSC 1	
8	Coarse tune 1	-60 .. 60 ST	0 ST	OSC 1	
9	Fine tune 1	-100 .. 100 cents	0 cents	OSC 1	
10	Harmonics 1	0 .. 127	64	OSC 1	
11	Timbre 1	0 .. 127	64	OSC 1	
12	Morph 1	0 .. 127	64	OSC 1	
13	FM 1	-100 .. 100 %	0 %	OSC 1	
14	Timbre mod 1	-100 .. 100 %	0 %	OSC 1	
15	Morph mod 1	-100 .. 100 %	0 %	OSC 1	
16	Low-pass gate 1	0 .. 127	127	OSC 1	
17	Time/decay 1	0 .. 127	64	OSC 1	
18	Model 2	0 .. 23	0	OSC 2	
19	Coarse tune 2	-60 .. 60 ST	0 ST	OSC 2	
20	Fine tune 2	-100 .. 100 cents	0 cents	OSC 2	
21	Harmonics 2	0 .. 127	64	OSC 2	
22	Timbre 2	0 .. 127	64	OSC 2	
23	Morph 2	0 .. 127	64	OSC 2	
24	FM 2	-100 .. 100 %	0 %	OSC 2	
25	Timbre mod 2	-100 .. 100 %	0 %	OSC 2	
26	Morph mod 2	-100 .. 100 %	0 %	OSC 2	
27	Low-pass gate 2	0 .. 127	127	OSC 2	
28	Time/decay 2	0 .. 127	64	OSC 2	
29	Model 3	0 .. 23	0	OSC 3	
30	Coarse tune 3	-60 .. 60 ST	0 ST	OSC 3	
31	Fine tune 3	-100 .. 100 cents	0 cents	OSC 3	
32	Harmonics 3	0 .. 127	64	OSC 3	
33	Timbre 3	0 .. 127	64	OSC 3	
34	Morph 3	0 .. 127	64	OSC 3	
35	FM 3	-100 .. 100 %	0 %	OSC 3	
36	Timbre mod 3	-100 .. 100 %	0 %	OSC 3	
37	Morph mod 3	-100 .. 100 %	0 %	OSC 3	
38	Low-pass gate 3	0 .. 127	127	OSC 3	
39	Time/decay 3	0 .. 127	64	OSC 3	
40	Model 4	0 .. 23	0	OSC 4	
41	Coarse tune 4	-60 .. 60 ST	0 ST	OSC 4	
42	Fine tune 4	-100 .. 100 cents	0 cents	OSC 4	
43	Harmonics 4	0 .. 127	64	OSC 4	
44	Timbre 4	0 .. 127	64	OSC 4	

#	Parameter	Range	Default	Page	Values / notes
45	Morph 4	0 .. 127	64	OSC 4	
46	FM 4	-100 .. 100 %	0 %	OSC 4	
47	Timbre mod 4	-100 .. 100 %	0 %	OSC 4	
48	Morph mod 4	-100 .. 100 %	0 %	OSC 4	
49	Low-pass gate 4	0 .. 127	127	OSC 4	
50	Time/decay 4	0 .. 127	64	OSC 4	
51	Output 1 gain	-40 .. 6 dB	0 dB	OUT	
52	Output 2 gain	-40 .. 6 dB	0 dB	OUT	
53	Output 3 gain	-40 .. 6 dB	0 dB	OUT	
54	Output 4 gain	-40 .. 6 dB	0 dB	OUT	
55	MIDI option 1	0 .. 144	141	MIDI	
56	MIDI option 2	0 .. 144	141	MIDI	
57	MIDI option 3	0 .. 144	141	MIDI	
58	MIDI option 4	0 .. 144	141	MIDI	
59	Share params 1	1 .. 4	1	MIDI	
60	Share params 2	1 .. 4	1	MIDI	
61	Share params 3	1 .. 4	1	MIDI	
62	Share params 4	1 .. 4	1	MIDI	
63	Bend range	0 .. 48 ST	2 ST	MIDI	
64	Envelope	0 .. 1	Trigger	OUT	0=Trigger, 1=Gate
65	Output mode	0 .. 1	Mixed	OUT	0=Mixed, 1=Individual
66	Output 1	0 .. 7	Osc 1 main	OUT	0=Osc 1 main, 1=Osc 1 aux, 2=Osc 2 main, 3=Osc 2 aux, 4=Osc 3 main, 5=Osc 3 aux, 6=Osc 4 main, 7=Osc 4 aux
67	Output 2	0 .. 7	Osc 2 main	OUT	0=Osc 1 main, 1=Osc 1 aux, 2=Osc 2 main, 3=Osc 2 aux, 4=Osc 3 main, 5=Osc 3 aux, 6=Osc 4 main, 7=Osc 4 aux
68	Output 3	0 .. 7	Osc 3 main	OUT	0=Osc 1 main, 1=Osc 1 aux, 2=Osc 2 main, 3=Osc 2 aux, 4=Osc 3 main, 5=Osc 3 aux, 6=Osc 4 main, 7=Osc 4 aux
69	Output 4	0 .. 7	Osc 4 main	OUT	0=Osc 1 main, 1=Osc 1 aux, 2=Osc 2 main, 3=Osc 2 aux, 4=Osc 3 main, 5=Osc 3 aux, 6=Osc 4 main, 7=Osc 4 aux

TROUBLESHOOTING

Problem	Fix
macOS says the app is damaged or from an unidentified developer; the DAW skips the plug-ins	The builds are not notarised. Run the `xattr -dr com.apple.quarantine …` command from the install chapter, then rescan plug-ins.
No sound from EX Multisample	Check the SD folder has sample folders (SD -> Show in Finder) and that a folder is selected in the left list.
Notes hang	Press PANIC (or MIDI -> All notes off).
An effect is silent	Granulator, Multi-FX, Resonator and Spectral Freeze need audio at the plug-in input. In the apps choose the input under Options -> Audio/MIDI settings and un-mute it. The Granulator also needs RECORD (or a loaded WAV) and a note or drone.
Logic does not list Granulator or Resonator as insert effects	They are MIDI-controlled effects: choose them in an instrument slot under AU MIDI-controlled Effects and select the audio as the side chain.
The DAW does not see Out 3/4	Enable the plug-in's second output bus in the DAW's routing/IO settings.
Auto-sampled files start late or early	Run TEST and set AS Latency just below the measured value.
Large folders take a moment to load	Samples load into memory in the background; very large libraries (hundreds of MB) take a few seconds.
A wavetable change does nothing	On the display, Wavetable needs confirming (PUSH V): press Esc, or choose the table in the list instead.

CREDITS AND LICENCE

- EX-GF is open source under the **GNU AGPL-3.0**; source code at github.com/femifleming/ex-gf.
- Built with JUCE 9 (AGPLv3). Poly FM, Macro Oscillator 2 and Resonator use the DSP code of Mutable Instruments Plaits and Rings by Émilie Gillet (MIT).
- Behaviour follows the disting EX user manual by Expert Sleepers. Not affiliated with Expert Sleepers.
- Sample library: VSCO-2 Community Edition, Karoryfer (Emily Guitar, Growlybass, Hadzi-Fia), Virtuosity Drums, AKWF wavetables, Greg Sullivan CP80 (CC BY 3.0) and generated content - see CREDITS.txt in the library for details.